

جامعة القصيم – كلية الهندسة

**Qassim University - College of
Engineering**



**Course First Day Material
Introduction to Engineering
Design-I GE 211
An Enhanced version**

(Active Learning Approach)

First Semester 1448 (2026/2027)

(Semester 481)

The 39th round of this course

Reviewed and updated by: **Engineering Design Professors Council**

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عزيزي الطالب

يسرنا أن نرحب بك في مقرر "مدخل إلى التصميم الهندسي-GE 211-I" ، فنحن شغوفون للتعرف عليك ولمساعدتك على بدء خطواتك على سلم "الهندسة" ، فهذه فرصة طيبة كي نقابلك ونمد لك يد العون لتنمية وتطوير قدراتك، إننا على يقين بأنه طالما أن سجلك ومعدلك الدراسي قد أهلك للالتحاق بكلية الهندسة فإن قدراتك تؤهلك لتكون مهندسا في المستقبل، ولكن هناك الكثير من الجهد الذي يجب أن تبذله لا لتكون مهندسا فقط بل لتكون مهندسا متميزا ومبدعا.

تم تصميم هذا المقرر من أجلك كي ينمى ويطور قدراتك كمهندس المستقبل، فنحن نهدف من مقرر "مدخل إلى التصميم الهندسي-I" إلى:

- تنمية وتطوير فهم الطالب لعملية التصميم الهندسي والتي تبدأ بتحديد الحاجات المطلوبة وتعريف هذه الحاجات تعريفا جيدا بالإضافة إلى المتطلبات الضرورية لذلك وصولا إلى استكمال المشروع المطلوب.
- وضع الطالب في بيئة تساعد على التفكير العلمي المبدع.
- توسيع فهم الطالب للمشاكل الهندسية ليشمل كل العوامل التي تؤثر على الحل النهائي للمشكلة.
- تزويد الطالب بالمعارف والمهارات التي تساعد على فهم طبيعة مهنة الهندسة.
- مساعدة الطالب على اكتساب مهارة وخبرة التصميم من خلال المشروع الهندسي.
- مساعدة الطالب على اكتساب مهارات العمل في فريق.
- مساعدة الطالب على قيادة فريق والقدرة على اتخاذ القرار بطريقة علمية.
- مساعدة الطالب على ممارسة أخلاقيات وضوابط مهنة الهندسة.
- مساعدة الطالب على تنمية وتطوير مهارات الاتصال من خلال تقديم العروض الشفهية والمكتوبة من خلال الأنشطة المختلفة للمقرر.

إننا نأمل انه مع انتهاء دراستك لهذا المقرر أن تكون قد كونت انطبعا ايجابيا عن مهنة الهندسة مع القدرة على فهم الكيفية التي يمكن بها حل المشاكل التي يواجهها المهندس بشكل يومي. كما أننا نتمنى ان تكون دراستك لهذا المقرر خير معين لك على التفوق في باقي المقررات الدراسية الأخرى. وفي النهاية نتمنى لك حياة دراسية مثمرة في كلية الهندسة مع التوفيق والنجاح الدائم. والسلام عليكم ورحمة الله وبركاته ...

مجلس أساتذة التصميم الهندسي

عنهم (أجدياً)

د/احمد الراجي - م/بشرى - د/حفصة - د/عبدالرحيم - د/محمد العصفوري - د/هاني داهش - د/هاني عمار

* ينصح بضرورة الاطلاع بصفة دورية على صفحة المقرر (المتاح من خلال موقع الكلية وموقع البلاك بورد) للوقوف على أحدث التنويهات

Getting Started

This course **packet** is given to each student on the first day of class. The packet contains starting materials for GE 211 as a whole (white sheets), as well as color-coded starting materials for the two individual parts of GE 211 - Concepts (**yellow**) and Laboratory (**orange**). Also included are a course Calendar (**pink**) and a Sign Off form (**blue**).

First-Day Package Table of Contents for Course Starting Materials (this part only)

1. Course structure - concepts and laboratory
2. Course contents
3. Classroom expectations - the need to participate
4. Professional and ethical behavior
5. Required course materials
6. Course celebration
7. Assessment of work and course grades
8. GE 211-pre-class-materials
9. Course fail situations

1. COURSE STRUCTURE – CONCEPTS AND LABORATORY

The course content and classroom activities have been divided into two subjects: **Concepts** and **Laboratory** (see Figure 1), which are alternating every other week. In this course, you will spend 6 contact hours per week in Concepts and then another six hours per week in Design Laboratory.

2. COURSE CONTENTS

GE 211 is a **required** introductory engineering course, which should be taken by all engineering students. The **Course Goal** is to introduce students to the Engineering Method, which is accomplished by focusing on five Course Objectives:

- Self-Regulation,
- Communication,
- Working Cooperatively and Collaboratively,
- Problem Solving, and
- Quality.

In addition, the following have been used to determine the course content:

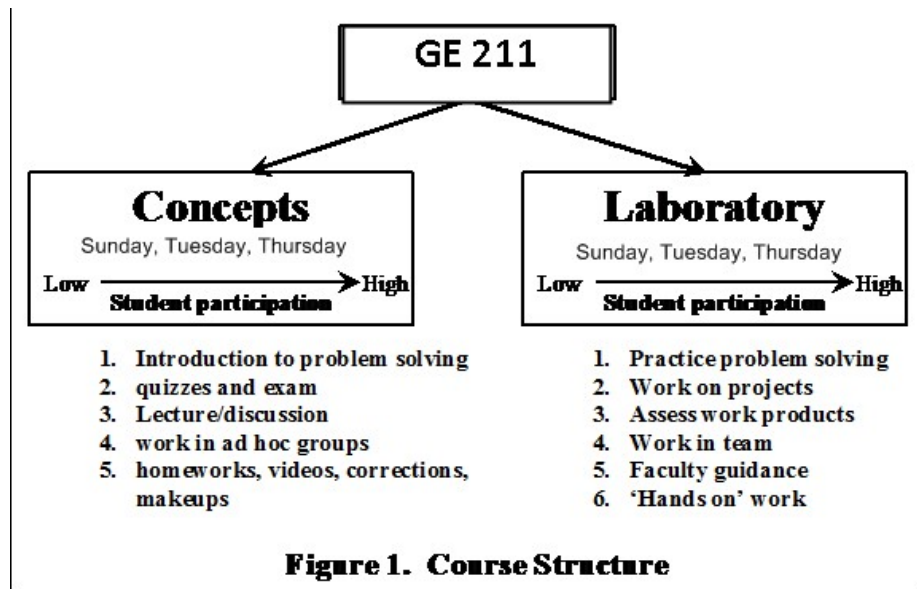
2.1. The course must be of use to all engineering disciplines.

We will focus on the **process of how to solve problems**.

In this course, we strive to develop your confidence in your ability to solve problems, regardless of their nature.

2.2. The course must develop knowledge that is useful in sophomore-level classes.

GE 211 will prepare you to meet the general **presentation requirements** associated with technical work (i.e., what is expected when you turn in a homework assignment). This will help you succeed with the discipline-specific problems that will come later.



2.3. The course should help students evaluate whether engineering is a profession of interest.

You can expect to:

- Use textbooks that present and discuss problems of the type engineers address,
- Receive general information about some specific engineering fields (e.g., what sort of problems a mechanical engineering graduate might confront),
- Learn problem-solving approaches usually used in industry,

3. CLASSROOM EXPECTATIONS - THE NEED TO PARTICIPATE

The course uses Active Learning procedures, student teams, and the concept of continuous improvement of the learning process¹. Constructivism² has also been used to develop several in-class activities. Thus, *you can expect to:*

- do a lot of talking while in class, both to small groups and to the entire class;
- be given problems that you think (at first) you do not know how to solve;
- find some of the classroom instructions not to be immediately clear;
- find that other students will have different interpretations of what is expected during class;
- be a team leader, team recorder, and team member at different times during the semester;
- help other students understand course material at the request of the course faculty;
- be regularly invited to suggest ways to improve the class;

¹ The material, in the Blue Workbook, is supplied to help you understand and function in an active, team-based classroom.

² A simple constructivist class exercise is organized as follows: teams are assigned a problem but are not given any suggestions on how the problem might be solved. The teams work until they are unable to proceed. At that time the instructor provides either some guidance or teams share (with other teams) their views on how to proceed. The teams then continue working on the problem and the cycle is repeated.

- be asked by a team member to modify the way you behave;
- ask a team member to modify the way he behaves, and
- be quite active during most class periods.

It has been the course faculty's experience that students who participate, even when they occasionally falter, have a much easier and more enjoyable time than students who resist participating. Your participation is a part of your course grade (see Section 5)

4. PROFESSIONAL AND ETHICAL BEHAVIOR

Professionalism is an approach and set of behaviors that all engineers are expected to exhibit as they pursue their profession. Some of these behaviors have been explicitly codified (e.g., Ethical Canons of Engineering), but other, less explicitly stated expectations should still be observed (e.g., use of appropriate language). An important part of your education is to become aware of, and to begin practicing, these professional behaviors.

GE 211 includes four specific behavioral expectations: Academic Integrity, Appropriate Language, Appropriate Classroom Behavior, and Appropriate Use of QU Facilities. Hospital/medical statement (or any out-of-class official activity)

4.1 Academic Integrity

The University requires each student to act with honesty and integrity and to respect the rights of others in carrying out all academic assignments. Several actions constitute a violation of the policy. These actions include, but are not limited to:

- referring to unauthorized materials or sources or employing unauthorized devices (e.g., audio recorders, calculators, or solution manuals) during tests, quizzes, homework, and class activities;
- possession, buying, selling, or otherwise obtaining or using, without appropriate authorization, a copy of any materials intended to be used for academic evaluation in advance of its administration;
- depending on the aid of others to the extent that the work is not representative of the student's abilities, knowing or having good reason to believe that this aid is not authorized by the instructor;
- submitting the ideas or work of another person or persons without customary and proper acknowledgment of sources (i.e., engaging in **plagiarism- the accepted ratio will be set the instructors**);
- permitting one's own ideas or work to be submitted by another person without the instructor's authorization.

University policy allows for cheating sanctions ranging from zero credit for an assignment to expulsion (without expectation of readmission) from the University. **Any student who is found to have violated the University's Academic Integrity Policy in GE 211, no matter how minor the violation, will have the sanctions set by the Engineering College Policy on cheating.**

4.2 Appropriate Language

Engineers are expected to communicate ideas effectively. Inappropriate language (written and oral) does not effectively communicate your ideas to an audience. In GE 211, you are expected to use appropriate language in:

1. all your written communications (assignments);
2. all your oral communications (presentations, conversations with peers or instructors); and
3. all your electronic communications (e-mail).

Use of inappropriate language will result in the issuance of a Self-Regulation Lapse; repeated use of inappropriate language will result in your withdrawal from the class.

4.3 Appropriate Classroom Behavior

In GE 211, you are expected to participate in the various classroom activities, including:

1. Coming/joining to each class on time;
2. Working on whatever assignment has been given;
3. In the case of online lecturing: Being connected and present to the online lecture all the time (*connection to the online class and stability of the internet is the student's responsibility and is out of any discussion, even though examinations. It is advised by the course professors to have an alternative means to get connected to the BB session in case your primary one failed*).
4. Being awake;
5. Not consuming any food or drink (except water) while in the QU classrooms.
6. Following instructions given by the instructor; and
7. Avoiding disruptive side conversations.

Repeated occurrences of inappropriate classroom behavior will result in a reduction in your grade (see "Self-Regulation Lapses" in Section 5)

4.4 Appropriate Use of QU Property and Facilities

In GE 211, you will use a variety of QU facilities and handle a variety of QU properties. You are expected to make appropriate use of these facilities and property, including:

1. gently handling **all items** distributed for your use and returning such items in the same condition as they were issued: folders, tools, handout sheets, etc.;
2. leaving a clean work space – tables, floors, and chairs;
3. Treating furniture gently – avoiding rough treatment, putting feet on tables and chairs, etc.

Especially serious abuse of QU property and facilities can result in university-level sanctions, including prosecution. Repeated (generally as few as five) minor occurrences of inappropriate use of QU property and facilities will result in a reduction in your grade (see "Self-Regulation Lapses" in Section 5).

4.5 Hospital/medical statement (or any out-of-class official activity)

One of the GE 211 bases is in-class practice, so student attendance is a must. Consequently, students are required to schedule any personal/medical/college/University ... activity to be away from the class scheduled time. If under any circumstances a student gets

sick (for example) and that prevents him from being in class; a written statement from QEC (not any other place) must be provided to the course professor council within two weeks from the occasion for approval. Upon the course professor council approval, the following constraints will be applied.

- Only **one** chance is given to a student per semester.
- The approval **removes** the absence penalty (1 lapse).
- If the missed work was an original **quiz**: The approval removes the quiz-NS penalty & allows the student to take this quiz with the same-quiz correction time (only) with no chance to get E. The approval, however, **does not allow** the student to make correction/makeup for this quiz.
- If the missed work was a **correction/makeup for certain quiz**: The approval allows the student to take it with the first scheduled time for any other correction/makeup even if he had another correction/makeup at that time.
- For any other work: the ordinary course norms are to be applied
- The approval **does not** affect any other penalty associated with late submission, class activity, and/or team uncompleted work.

5. REQUIRED COURSE MATERIALS

The following is a list of items that you (or your team) are required to purchase.

1. *Introduction to Engineering Design, The Workbook (a.k.a., The Blue Workbook)*, Barry McNeill, Lynn Bellamy, and Veronica Burrows
2. *Strategies For Creative Problem Solving*, H. Scott Fogler and Steven E. LeBlanc
3. A package of Post-it or note cards (3' x 5" is a good size)
4. One folder, a set of dividers, and a package of transparency sheets

6. CELEBRATION

There is a Final Exam in GE 211, and it is called The Celebration. The official date/time for the Celebration is announced on the college calendar and/or the on PART 3 of the FDM. Any group has the right to change this time/date subject to 100% approval of the group- students and the course instructor. The celebration must be attended and practiced by each student in the existence of all his-group students; otherwise, **the student fails the course.** (read item 9 for more details)

7. ASSESSMENT OF WORK AND COURSE GRADES

This course always follows the rules of the continuous improvement process. That is why, *and based on the student feed backs from last semesters*; a new process in submitting work will be utilized. That is some of the home works and assignments (that is produced by the MS-office) must be uploaded to the **Black Board**. BB-Links will be available till the start of the due lecture. Any checklist and/or handworks must be submitted in class as hardcopies.

Most students find that assessment (grading) is the area in which they find the most discrepancy between their **expectations** of the course and the **reality** of the course. We wish to reassure students that the GE 211 assessment process is carefully and rationally designed - it is organized to help students define and recognize **quality**, and to motivate students to achieve **quality work**. The following material will discuss how the course assesses your **cognitive** behavior, and your **affective** behavior

and how these two assessments are combined to generate a grade.

7.1 Assessment of Cognitive Behavior (What you Know)

The process used to assess all of your work is discussed in Section B of the Blue Workbook (with some modifications to be suitable for the online classrooms if applicable). Your grade for a work product is based on the **quality of work** you or your team submit. Quality work is defined as essentially complete and correct work that is documented neatly in a well-organized fashion and submitted on time. To help you submit quality work, in each Session, you will be referred to a set of **Checklists** that define what is expected for your assignments.

Submitted work is assessed either as Exceeding Expectations (**E**), Meeting Expectations (**M**), Needing Improvement (**NI**), or No Credible Effort (**NCE**)/No Submittal (**NS**).

- To **meet expectations (M)** on any given assignment, you must meet **all** expectations on that assignment's checklist. Failure to meet expectations for **any** item on the checklist, will result in a grade of **needs improvement (NI)**.
- To **exceed expectations (E)**, the work must **first meet expectations** and then, in addition, exhibit some relevant work beyond what was requested in the assignment. Elements of exceeding expectations might include demonstrating *a level of understanding* beyond that asked for in the assignment description, answering a *relevant question* not originally asked for, or doing some extraordinary level of achievement.
- If work is assessed as **NI**, you are given another chance to make improvements and resubmit the work (this is applied for some works only – review your FDM). Historically, provided a credible effort is made to improve the work, in general, all resubmitted work is reassessed as meeting expectations (**M**). Failure to improve an NI harms your grade.
- Work that does not demonstrate at least a credible effort towards completion is assessed as no credible effort (**NCE**) and a *Self-Regulation Lapse* is recorded. When work is not submitted, an assessment of No Submission (**NS**) is issued, and a *Self-Regulation Lapse* is also recorded. You are not allowed to submit corrections for some specific works. For some other works, however; you are given another chance to submit **make-up work**. Work that is not made up (i.e., retains the **NCE** or **NS**) has a severe negative impact on the session grade.
- Be aware of plagiarizing someone else work. The plagiarism will be checked for materials submitted with the current and previous semesters. Usually this is done for the e-submission of the works. This semester; some works are required to be uploaded to the BB. Some others are not required. However; for these assignments which are not required to be uploaded to the BB; the instructor has the right to ask any team to upload any of his work to the BB to check the plagiarism.
- Each student need to make special record for himself/herself about the assignment grades (as he/she is having all his/her work), absence, lateness, bonuses and lapses. Therefore; each student can expect his/her final grade in GE211. On the other hand, only three items are not subject of announcement (however, a student will be notified how to grade them in advance): The results of the defect removal + the grades of the final assessment of the DNB and the grades of the final presentation on the celebration day.

7.2 Assessment of Affective Behavior (Self-Regulation)

Independent of work quality, affective *behavior* (as described in Section 4) also impacts your GE 211 grade. One of the course goals is that students demonstrate *self-regulation*³. Instances of poor behavior or participation are recorded as *Self-Regulation Lapses*. *Self-Regulation Lapses* include, but are not limited to:

- Lateness to class (1/2 Lapse);
- Absences from class⁴ (1 Lapse);
- Being connected to the BB-lecture but not physically present (1 Lapse) in case of online lecturing;
- Using mobiles (1 Lapse);
- Asking for permission to go out of class for a few minutes while you are not in need to (1 Lapse);
- Being unprepared for class (1 Lapse);
- Failure to participate and/or disruptive or unethical classroom behaviors, (e.g., side conversations, sleeping, working on homework assignments for other classes, etc.) (see Section 4 of the First Day handout) (1 Lapse);
- Minor inappropriate use of QU properties and facilities, (1/2 Lapse);
- In case of **permitted resubmission**: late submittal: (upon approval- of an assignment) 1 Lapse if submitted late within the first week - *No re-submission is accepted after one week of its due date*); and
- Getting (NCE) on an assignment for the 1st time which is changed to meet later (1 Lapse).

It may be appropriate at this point to mention that the course instructor may grant a bonus to one or more of his class students based on the quality of a specific work that is out of assessment. A student cannot be granted more than one bonus per class. A single bonus can remove the effect of half lapse only. GE 211 has nothing to do with any bonuses given by the college.

7.3 Session Grades - The General Philosophy

Session grades are based on the **final** quality of the assignments you submit⁵. If by the end of the semester, you have not improved any NI's, NCE's, and/or NS's you received during the semester or if you have accumulated more than four Self-regulation lapses you will accumulate **Defects** that harm your grade⁶. How the final assessed quality (E, M, NI, NCE, NS, Defects) is converted to a session grade can be found in tables located in the first-day materials for each session (see colored pages). There are several important aspects to determining your Session grades. In particular:

1. To get an **A**, at least **FOUR** assignments must Exceed Expectations in the Concept session or **FIVE** Exceed Expectations in the Lab session, and there are **no Defects in each one**.
2. To get a **B**, all assignments must at least Meet Expectations, and there are **no Defects**.
3. To get a **D** or **F** in a session, you or your team must have demonstrated a significant *lack* of effort (e.g., one assignment never done or not done at a credible effort) or lack of self-regulation (e.g. work turned in late all semester).

³ See Section B of the Blue Workbook for more information on Self-Regulation.

⁴ Since the student participation is a **MUST**, the **excused-absence** is applied for the student absence percentage only and is not applied to the course-grading-sheet.

⁵ Work that is **initially** assessed as NI, NCE or NS but is assessed as M on **resubmission** is treated as **M** in determining grades.

⁶ Because students sometimes accumulate too many Self-regulation lapses, usually in the first week C1, before they fully understand the consequences, students are provided an opportunity to remove **ONE** Defect in Concepts.

7.4 Course Grade

The conversion of Session grades into course grades is based on your pattern of performance in the two Sessions (Concepts and Lab). Table 1 shows the course grade assigned for every possible combination of sessions. (NB: **Whoever gets 5 defects** in concept or lab session will get an "F" in the course, also, whoever gets 4 defects in concept or Lab session will never exceed "D" as the course final grade.)

Table 1. Course Grade for Every Possible Pattern of Session Grades (*except the cases listed above*)

Session grades	Course grade	Session grades	Course grade
A, A	A+	B, D	C+
A, B	A	B, F	C
A, C	B+	C, C	C+
A, D	B	C, D	C
A, F	C+	C, F	D+
B, B	B+	D, D	D+
B, C	B	D, F	D

8. GE 211-PRE-CLASS-MATERIALS

All of the documents used in GE 211, will be available (*before the associated classes by a suitable time*) for previewing/printing from the Black-Board system. Most of the documents are MS Office files but several of the documents have PDF extensions which means that they require the Adobe Acrobat Reader to view and print.

You will need to print/reuse all the checklists and other materials that are used in the course. Be notified that most checklists require one copy per student; some may require one copy for the entire team. So, before you print anything ask yourself:

- "Why do I need a hard copy of this document?"
- "Will I likely keep this material past the end of this week, or the end of this semester?"
- "Is this copy required for each member or one copy of it is enough for the team?"

9. COURSE FAIL SITUATIONS: (include but are not limited to)

- Having a grade of F in both sessions of the course
- Accumulates **FIVE** defects or more in **one** of the course sessions (**whatever** the grade of the other course session).
- Neither attending nor practicing the celebration at its scheduled and announced time (there is no replacement of this celebration day if considered as final exam)
- Violation of QU-Academic-Integrity policy (with a recommendation from the course professors council)
- Violating the attendance ratio according to the college and/or the university policies.

Student Sign Off

Students who are not comfortable signing this document must request a meeting with the course coordinator before the third week of the semester to review the requirements as necessary

Name: _____

Date: _____

Group # _____

Team # _____

Member # _____

I have read and understand all of the First Day Materials for the Course as well as the First Day Materials for the two parts of the course (Concepts and Lab). *In particular, this includes:*

1. I am responsible for keeping up-to-date with announcements and other course materials;
2. I am expected to fulfill all of the expectations and behavior requirements described in the sections titled Course Content, Classroom Expectations, and Professional and Ethical Behavior;
3. I am expected to come ready for class which includes being there on or before time, printing all the required class materials before class, and bringing these materials to class.
4. If my behavior in GE 211 is deemed significantly unprofessional (*e.g.*, using inappropriate language), I will be withdrawn from the course.
5. If I am caught cheating/ **plagiarism some-one-else-work**, I know that:
 - a) the consequences will be set by the Engineering College's Policy on cheating,
 - b) the sanctions for cheating could be as high as expulsion with a grade of Exceed, failure for cheating, entered on my transcript, and
 - c) cheating includes *but is not limited to*:
 - i) Submitting work that is not my own or, for team assignments, not my team's.
 - ii) Accepting unauthorized help from other students **or** providing unauthorized help to other students (for example, giving another student a copy of the work you plan to turn in for a homework assignment);
 - iii) Using unauthorized materials during quizzes or copying someone else solution(s); and
 - iv) Supply any of my course-mate with solution(s) of any quiz questions; and
 - v) For team assignments, accepting a grade or other credit for teamwork to which I have not made an appropriate contribution.

Signature



[An electronic copy of this page is available on the blackboard (GE 211) link that must be submitted no later than the announced Session. Students who are not comfortable signing this document must request a meeting (virtually) with the course coordinator before the third week of the semester to review the requirements as necessary.